

Trace and Rank of Real Idempotent Matrices

The MerLean research agent*

Abstract

A real square matrix A with $A^2 = A$ has trace equal to its rank; in particular its trace is a nonnegative integer. The proof factors $A = BC$ with the rank installed as the inner dimension and shows $CB = I_r$, reducing the claim to a trace computation. The identity holds over every field in the characteristic-free form $\text{tr } A = (\text{rank } A) \cdot 1_F$; characteristic zero is used only to read the field element $(\text{rank } A) \cdot 1_F$ as the integer $\text{rank } A$, and the order on $\mathbb{R}$ only for the bound $0 \leq \text{tr } A \leq n$. The argument was developed and audited by an adversarial proof-ledger process, and the displayed identity of the theorem is checked end to end by a proof-assistant kernel under the standard axioms — by an independent route: the factorization proof printed here is itself formalized only at its crux. The verification model is described in Section 4.

1 Introduction

Idempotent matrices — the matrices of (generally oblique) projections — satisfy a striking rigidity: the trace, a priori an arbitrary real number, always equals the rank. This note records a complete proof of that fact — self-contained modulo standard finite-dimensional linear algebra (existence of a basis, dimension of a span) — together with its exact scope in terms of the underlying field.

This is a demonstration artifact. The result is a textbook exercise; what the note exhibits is a pipeline that takes an informally stated question to an adversarially audited proof with a machine-checked certificate at its crux, and then to this document. The reader interested in the pipeline rather than the mathematics should read Section 4 first.

Section 2 fixes notation. Section 3 proves the result: four lemmas, the characteristic-free proposition, and the theorem — of which Lemma 3 is the crux — followed by a remark delimiting the scope. Section 5 states what a full formalization would add.

2 Setup

Throughout, F is a field and $M_{p \times q}(F)$ denotes the $p \times q$ matrices over F , with $M_n(F) := M_{n \times n}(F)$; matrices may be empty ($p = 0$ or $q = 0$), a product over an empty inner index is the zero matrix (each entry an empty sum), and $I_r \in M_r(F)$ is the identity. For $M \in M_n(F)$ the *trace* is $\text{tr } M := \sum_{i=1}^n M_{ii}$ (zero for $n = 0$). For $M \in M_{p \times q}(F)$, the *column space* is

*A demonstration artifact: this note was produced end-to-end by the MerLean pipeline (route portfolio, adversarial step-ledger debate, targeted kernel-checked certification, and the gated paper-writing protocol) from the campaign record in the MerLean repository, <https://github.com/MerLeanProver/MerLean>, directory `examples/lite-research-DEMO-IdempotentTraceRank`. The mathematics is standard undergraduate material and no novelty is claimed for it.

$\text{col } M := \{Mx : x \in F^q\} \subseteq F^p$ — equivalently, since $Mx = \sum_j x_j(Me_j)$, the span of the columns — and the *rank* is $\text{rank } M := \dim_F \text{col } M$, a nonnegative integer at most $\min(p, q)$. Finally $\iota : \mathbb{Z} \rightarrow F$ is the unique ring homomorphism, $\iota(m) = m \cdot 1_F$; “the field element x is the integer m ” means $x = \iota(m)$. A matrix $A \in M_n(F)$ is *idempotent* if $A^2 = A$.

3 The factorization argument

The four lemmas hold over every field F ; no property of $\mathbb{R}$ enters before Theorem 6.

Lemma 1 (rank factorization). *Let $A \in M_n(F)$ have rank r . Then there exist $B \in M_{n \times r}(F)$ and $C \in M_{r \times n}(F)$ such that $A = BC$ and $\text{col } B = \text{col } A$; in particular $\text{rank } B = r$.*

Proof. For $r = 0$ take the empty matrices ($A = 0$ because every column of A lies in the zero-dimensional $\text{col } A$; both column spaces are $\{0\}$). For $r \geq 1$, choose a basis $b_1, \dots, b_r$ of $\text{col } A$ and let $B := [b_1 \cdots b_r]$; its columns being a basis, $\text{col } B = \text{col } A$ and $\text{rank } B = r$. Each column Ae_j has unique coordinates $c_{1j}, \dots, c_{rj}$ in this basis; let $C := (c_{kj})$. Then $(BC)e_j = B(Ce_j) = \sum_k c_{kj}b_k = Ae_j$ for every j , so $A = BC$. $\square$

Lemma 2 (cancellation). *Let $B \in M_{n \times r}(F)$ have rank r . Then for all $X, Y \in M_{r \times m}(F)$: if $BX = BY$ then $X = Y$.*

Proof. If $Bv = 0$ for some $v \neq 0$, say $v_k \neq 0$, then b_k is a linear combination of the other columns, so $\text{col } B$ is spanned by $r - 1$ vectors, contradicting $\text{rank } B = r$; hence $Bv = 0$ forces $v = 0$. Now $B(X - Y) = 0$ makes every column of $X - Y$ such a v . $\square$

Lemma 3 (the crux). *Let $A \in M_n(F)$ be idempotent of rank r , and let $A = BC$ be a factorization as in Lemma 1. Then $AB = B$ and $CB = I_r$.*

Proof. For $v \in \text{col } A$, say $v = Aw$, idempotence gives $Av = A^2w = Aw = v$: the matrix A fixes its column space pointwise. Each column Be_j lies in $\text{col } B = \text{col } A$ (Lemma 1; for $r = 0$ there is nothing to prove), so $(AB)e_j = A(Be_j) = Be_j$ for all j , i.e. $AB = B$. Then

$$B(CB) = (BC)B = AB = B = BI_r,$$

and Lemma 2 cancels B on the left: $CB = I_r$. $\square$

Lemma 4 (trace commutation). *For $M \in M_{n \times r}(F)$ and $N \in M_{r \times n}(F)$: $\text{tr}(MN) = \text{tr}(NM)$.*

Proof. Both sides equal the finite sum $\sum_{i=1}^n \sum_{k=1}^r M_{ik}N_{ki}$, by expanding the diagonal entries and exchanging the two finite summations; for $n = 0$ or $r = 0$ both sides are 0. $\square$

Proposition 5 (characteristic-free form). *Let F be any field and $A \in M_n(F)$ idempotent. Then $\text{tr } A = \iota(\text{rank } A)$ in F .*

Proof. Let $r := \text{rank } A$ and factor $A = BC$ by Lemma 1. By Lemmas 3 and 4,

$$\text{tr } A = \text{tr}(BC) = \text{tr}(CB) = \text{tr}(I_r) = r \cdot 1_F = \iota(r). \quad \square$$

Theorem 6. *Let $n \geq 0$ be an integer and let $A \in M_n(\mathbb{R})$ satisfy $A^2 = A$. Then*

$$\text{tr } A = \text{rank } A,$$

i.e. $\text{tr } A = \iota(\text{rank } A)$; in particular $\text{tr } A$ is a nonnegative integer, uniquely equal to $\text{rank } A$, and $0 \leq \text{tr } A \leq n$.

Proof. Proposition 5 over $F = \mathbb{R}$ gives $\text{tr } A = \iota(r)$ with $r = \text{rank } A$. Since $\mathbb{R}$ is an ordered field, $m \cdot 1_{\mathbb{R}} > 0$ for every integer $m \geq 1$; hence ι is injective (so the integer with $\iota(m) = \text{tr } A$ is unique, and it is $r \geq 0$) and order-preserving (so $0 \leq r \leq n$ gives $\iota(0) \leq \iota(r) \leq \iota(n)$, i.e. $0 \leq \text{tr } A \leq n$). $\square$

Remark 7 (exact scope). *Idempotence is consumed exactly once (the first sentence of Lemma 3’s proof), and the passage from Proposition 5 to Theorem 6 is the only step that uses a property of $\mathbb{R}$ beyond the field axioms alone — its characteristic and, for the bound, its order. Over $F = \mathbb{F}_p$ the proposition still holds, and the theorem’s reading genuinely fails: $A = I_p$ is idempotent with $\text{tr } A = p \cdot 1 = 0$ while $\text{rank } A = p$. What breaks first is the injectivity of ι (and the order clause has no analogue over $\mathbb{F}_p$ at all). No symmetry, orthogonality, or diagonalizability of A is assumed anywhere; oblique idempotents are covered without change.*

4 Verification

The proof above was produced and audited as a 19-row dependency ledger: one claim per row with its exact prerequisites, over a conventions row fixed once, developed by one automated prover role and attacked over two rounds by an independent automated referee role. Round one raised one substantive objection — a citation whose printed location could not be verified, resolved by replacing the citation with the self-contained construction now inside Lemma 1 — and the ledger closed in round two with no fatal or gap-level objections, after a whole-graph audit of acyclicity and row interfaces; its eight remaining nitpicks — two of them regressions introduced by the round-one repairs — were applied after closure. The factorization chain was replayed numerically on 400 random idempotent instances, with a 200-instance deliberately oblique rerun and a 500-pair test of the trace commutation including the degenerate shapes $n = 0$ and $r = 0$; the identity itself was separately checked at $n = 0$, $A = 0$, and $A = I$ — about 1,100 instances in the deposited probes in total, plus a 4,000-instance referee screen recorded in the debate report. A necessity probe confirmed that dropping idempotence breaks $CB = I_r$ on every sampled instance. The crux inference — $CB = I_r$ from $A = BC$, $AB = B$, and cancellation, i.e. Lemma 3’s second half — was additionally checked by a proof-assistant kernel as a standalone certificate whose hypotheses are the prior ledger rows, with the standard three-axiom audit and nothing else. Finally, the displayed identity of Theorem 6 is formalized and kernel-checked end to end, with no unproved hypotheses and exactly the standard three axioms in its audit (the theorem’s “in particular” consequences are immediate from the identity and are not separately formalized); the formal proof reaches the statement through the proof assistant library’s projection-trace machinery — structurally the spectral viewpoint of the route this campaign retired, rather than the factorization route printed above — so discovery and certification took legitimately different roads, and both are part of the record. The full campaign record — the deposited experiment scripts, the debate conclusions, and both certified files with their audits — is in the repository cited on the title page.

5 Outlook

Two items remain open on the formal side, neither affecting the theorem’s status. First, the kernel-checked proof certifies Theorem 6 through the library’s projection machinery; formalizing the factorization route printed in Section 3 — the rank factorization of Lemma 1, including its

$\text{col } B = \text{col } A$ clause, and the independence argument of Lemma 2 — would make the discovery-side proof itself machine-checked, not only its conclusion and its crux. Second, Proposition 5, the characteristic-free form over an arbitrary field, is stated and proved here but was not formalized; it is the natural next anchor. The mathematical content of this note is complete as stated, and Remark 7 delimits the theorem exactly.